

A Packing of 7 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.969521. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.908876, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.969521 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.908876, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 7/6 = 2.969697$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N7.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
424217361145890604960221906188985001236407988599378121080222269304409557
186619648302845701989974818440432384698666914975740069642493229365293011
656612294346997057054355387250817981096720896771
```

denominator:

```
142857142857678898691998182354601901129743616044660546273045343399108661
911936246465135514863423705714240784616945812143202025654783484206923848
916013882180030113564860526168348280281769585914
```

B Coordinates

7 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N7.txt`.

```
25108430366 -591122335368 304115776843 104306063447128407024745 -182486765689 -87780895799 -2741663118250717935551 -20004545610 -338968709460 198828222916
-145571831258 -41028865353 -207245495198 -10866580812335606721 -247166074794 -3405823177 2699235696621360851426 103022117755 -401168924983 -452176609716
281465589413 59665447052 -152136519171 3595571759788245815910 309818654307 149214006544 193259046383 -159665781226 -92299819173 -134402676919 564160279088
430570725112 -287990878156 233956470045 135505223156136092660 -343897110405 -178015306278 -671296589677518875683 -240284108633 -397736165947 97308889965
290059239501 355384764236 -65469767485 -61843938394870430371005 246757305762 -53689080611 -3267056125050450856404 62603790001 223832074810 249277641725
-390541689934 -26683501377 -64330358304 41229849262-42445540733 163316116209 505797623412 -184571809308532700893 -132623995779 -493138038305 -269734606996
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2464668069 21245544792 174164640414 14319109120 476594075487 147700219287 -196080282726 3522399080333161314641 -328739788593 25219645283 301491455067
-73221725199 99040442905 531928823271 -41972403250420697142334 322923257288 563957583154 25859400731400020236977 -366369351193 155297010547 -200179797981
299252700918 78813930915 -194602830532 -13903787076815818421560 -164737763377 53394114311 22268275903311169523464 -395471357019 -185544831260 386219067719
320234012262 -445430408849 -145163760746 -39041190633260712115 -397257218120 129990612328 -393648055489593359881 267731881459 -100569265335 716106719658
366586645088 332820627299 -413518524714 -131302878389961908852 31427635120 516561534777 -9262759746 -101421615249 -234279495738 492747871689 166565999839
-75139156005 513130175798 -45626470722 -52552563622817873187057 142017743018 -264934572681 14275656205519745092990 -25489279460 -446491175300 288069866210
119992494591 -242465905705 -90564828735 -4286925640154760086435 -70040748185 -92216835690 697797459942899291382 79818357406 -15825629211 -525312187303
258094562950 100200452691 -257604591177 -3640833217359215159570 123384041645 -4264911991576 -802032527672703843050 -33434583532 291426003916 -434968105986
512297701465 49230567273 547476970178 239926149749 -391489624848 612299245027 -208937706182 -2309597494834212821048 -224095516013 377039751792 346481695515
31372067339 -308171852616 253466697086 -3963918327089138721221 -495396906653 505415775277 16650033389295658118093 -440134065582 -433815649395 55832209725
-118231478330 479049792939 235825167785 -243516604439150405821 -667332164240 -139728506975 -494588084203812089011 -571675939828 171018529338 -130107971946
-210669147096 -119546265066 -517668789386 2310733757301231096320 158224298707 202308776710 30906021757-184186490111 -192384665753 -51212074165 188561031135
514841744168 106322945040 -76451798740 19851511262751666419215 -15451899222 -455388501517 15710208941916209513045 89957887788 568333800715 224671357531
-249771761627 -330274489661 391103852699 4482735503709284262682 258856785059 103886436539 -130668559408959981519 -153415213480 -80933284420 -382979015502
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163805982669 338832783981 -588389074629 -58047069858286812943987 2055343670 -292564871966 45075071335 -448941435682 -107098022720 -40214242051 -280644868862
-38020099096 -517229742671 6297531297 77561702796 144891242547 238236713261 258873698726 -335486447362
```